

TrES-5b

R-1967

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_190916 · created 2026-08-06T23:26:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458742.7384 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.048 +/- 0.053
Transit depth (Rp/Rs)^2	0.23 +/- 0.5 [%]
Orbital Inclination (inc)	79.77 +/- 2.91
Airmass coefficient 1 (a1)	0.035 +/- 0.02
Airmass coefficient 2 (a2)	3.0 +/- 1.6
Scatter in the residuals of the lightcurve fit is	61.68 %
Best Comparison Star	None
Optimal Aperture	1.11
Optimal Annulus	6.5
Transit Duration (day)	0.013 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.048 ± 0.053 vs 0.142 (deviation 1.77σ)
Duration fit vs published	0.013 d vs 0.0758 d (deviation 4.19σ)
Mid-time O–C	6.1 min
Depth SNR	0.9
Residual scatter	61.68 %

Light curve

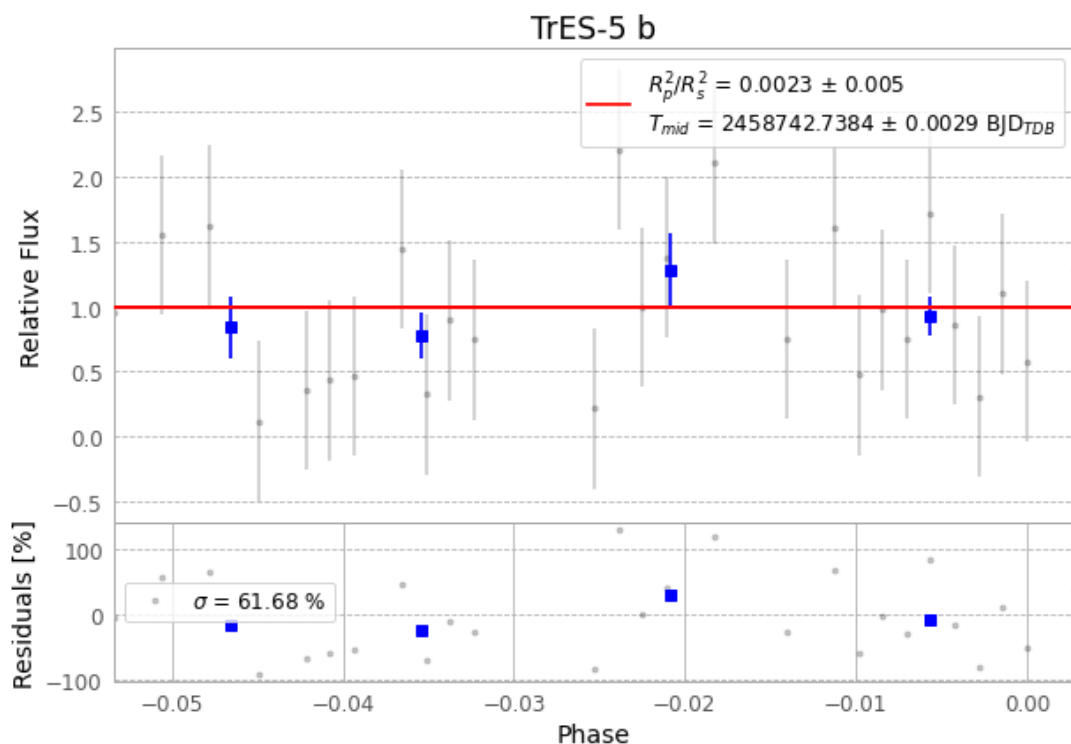


Figure 1 — FinalLightCurve_TrES-5 b_2019-09-15.png (SHA-256 08dd93f96e7810aa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [73, 309], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 31024d41e499d719...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

77 FITS frames (51 MB), TRES-5190916033612.FITS → TRES-5190916072412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-190916.log (695d7f995789...), FinalLightCurve_TrES-5 b_2019-09-15.png (08dd93f96e78...), FinalLightCurve_TrES-5 b_2019-09-15.csv (157c10c7382a...)

Compute resources

Wall time 102.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 23:26Z.